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Course: Introduction to Probability and Statistics

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Exercise 1

Let X_1 and X_2 be independent exponentially distributed random variables with parameters $1/\lambda_1$ and $1/\lambda_2$ respectively. Then the p.d.f. of X_1 is given as

$$f_{X_1}(x) = \lambda_1 e^{-x\lambda_1} \quad (1)$$

and the p.d.f. of X_2 is given as

$$f_{X_2}(x) = \lambda_2 e^{-x\lambda_2} \quad (2)$$

Define random variables N , U , V , and W as

$$N = \begin{cases} 1 & \text{if } X_1 < X_2 \\ 2 & \text{if } X_1 \geq X_2 \end{cases} \quad U = \min \{X_1, X_2\}, \quad V = \max \{X_1, X_2\}, \quad W = |X_1 - X_2|$$

Since X_1 and X_2 are i.i.d random variables, then we can express their joint probability distribution as a product of each of their individual probability density functions. That is

$$f_{X_1, X_2}(x, y) = \lambda_1 \lambda_2 e^{-(x\lambda_1 + y\lambda_2)}$$

1. We want to show that $P(N = 1) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$ and $P(N = 2) = \frac{\lambda_2}{\lambda_1 + \lambda_2}$

Consider the event where $N = 1$ then we know that $X_1 < X_2$ and thus

$$P(N = 1) = P(X_1 < X_2).$$

Notice that fixing a value for X_2 , say y_0 then we have the following conditional probability:

$$\begin{aligned} P(X_1 < X_2 | X_2 = y_0) &= P(X_1 < y_0 | X_2 = y_0) \\ &= \int_0^{y_0} f_{X_1}(x) dx && \text{Since } X_1 \text{ and } X_2 \text{ are independent} \\ &= \int_0^{y_0} \lambda_1 e^{-x\lambda_1} dx && (3) \\ &= -e^{-x\lambda_1} \Big|_0^{y_0} \\ &= 1 - e^{-y_0\lambda_1} \end{aligned}$$

But since we fixed a value for X_2 then we account for all possibilities of X_2 thus we have that

$$\begin{aligned}
P(X_1 < X_2) &= P(X_1 < y \mid X_2 = y) \quad \text{for all possible values of } X_2 \\
&= \int_0^\infty P(X_1 < y \mid X_2 = y) dy \\
&= \int_0^\infty P(X_1 < y) f_{X_2}(y) dy \\
&= \int_0^\infty (1 - e^{-y\lambda_1}) (\lambda_2 e^{-y\lambda_2}) dy \\
&= \int_0^\infty (\lambda_2 e^{-y\lambda_2}) dy - \int_0^\infty (\lambda_2 e^{-y(\lambda_2 + \lambda_1)}) dy \\
&= -e^{-(y\lambda_2)} \Big|_0^\infty + \lambda_2 \left(\frac{-1}{\lambda_1 + \lambda_2} e^{-y(\lambda_1 + \lambda_2)} \right) \Big|_0^\infty \\
&= 1 - \frac{\lambda_2}{\lambda_1 + \lambda_2} \\
&= \frac{\lambda_1}{\lambda_1 + \lambda_2}
\end{aligned}$$

for $P(N = 2)$ notice that the event $N = 2 \iff X_1 \geq X_2$ thus

$$P(N = 2) = P(X_1 \geq X_2) = 1 - P(X_1 < X_2) = 1 - \frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{\lambda_2}{\lambda_1 + \lambda_2}$$

2. Since $U = \min\{X_1, X_2\}$ then the event that $U > t$ means that $X_1 > t$ and $X_2 > t$, thus the intersection of these two events. Now since X_1 and X_2 are independent random variables then we have that

$$\begin{aligned}
P(U > t) &= P(X_1 > t)P(X_2 > t) \\
&= (1 - P(X_1 < t))(1 - P(X_2 < t)) \\
&= 1 - P(X_1 < t) - P(X_2 < t) + P(X_1 < t)P(X_2 < t)
\end{aligned} \tag{4}$$

Notice from equation 3 that $P(X_i < x) = 1 - e^{-x\lambda_i}$ for $i \in \{1, 2\}$ Thus we have that from equation 4 we get

$$\begin{aligned}
P(U > t) &= 1 - (1 - e^{-t\lambda_1}) - (1 - e^{-t\lambda_2}) + (1 - e^{-t\lambda_1})(1 - e^{-t\lambda_2}) \\
&= e^{-t(\lambda_1 + \lambda_2)}
\end{aligned}$$

3. We want to show that

$$P(N = 1, U > t) = P(N = 1)P(U > t) \quad \text{and} \quad P(N = 2, U > t) = P(N = 2)P(U > t).$$

For $P(N = 1, U > t)$ we have that

$$P(N = 1, U > t) = P(X_1 < X_2, \min\{X_1, X_2\} > t) = P(t < X_1 < X_2).$$

thus,

$$\begin{aligned}
P(t < X_1 < X_2) &= \int_t^\infty \int_t^y \lambda_1 e^{-\lambda_1 x} \lambda_2 e^{-\lambda_2 y} dx dy. \\
&= \int_t^\infty \lambda_2 e^{-\lambda_2 y} \left[\int_t^y \lambda_1 e^{-\lambda_1 x} dx \right] dy \\
&= \int_t^\infty \lambda_2 e^{-\lambda_2 y} \left[-e^{-\lambda_1 x} \right]_t^y dy \\
&= \int_t^\infty \lambda_2 e^{-\lambda_2 y} (e^{-\lambda_1 t} - e^{-\lambda_1 y}) dy \\
&= e^{-\lambda_1 t} \int_t^\infty \lambda_2 e^{-\lambda_2 y} dy - \int_t^\infty \lambda_2 e^{-(\lambda_1 + \lambda_2)y} dy \\
&= e^{-\lambda_1 t} \left[-e^{-\lambda_2 y} \right]_t^\infty - \left[-\frac{\lambda_2}{\lambda_1 + \lambda_2} e^{-(\lambda_1 + \lambda_2)y} \right]_t^\infty \\
&= e^{-\lambda_1 t} e^{-\lambda_2 t} - \frac{\lambda_2}{\lambda_1 + \lambda_2} e^{-(\lambda_1 + \lambda_2)t} \\
&= e^{-(\lambda_1 + \lambda_2)t} \frac{\lambda_1}{\lambda_1 + \lambda_2}.
\end{aligned}$$

Hence,

$$P(N = 1, U > t) = \frac{\lambda_1}{\lambda_1 + \lambda_2} e^{-(\lambda_1 + \lambda_2)t} = P(N = 1)P(U > t)$$

Notice that, the event $N = 2$ and $U > t$ is equivalent to $X_1 \geq X_2$, and $\min\{X_1, X_2\} > t$. That is

$$X_1 \geq X_2 > t$$

hence we have the following probability:

$$P(N = 2, U > t) = P(X_1 \geq X_2 > t).$$

That is,

$$\begin{aligned}
P(t < X_2 \leq X_1) &= \int_t^\infty \int_t^x \lambda_1 e^{-\lambda_1 x} \lambda_2 e^{-\lambda_2 y} dy dx. \\
&= \int_t^\infty \lambda_1 e^{-\lambda_1 x} \left[\int_t^x \lambda_2 e^{-\lambda_2 y} dy \right] dx \\
&= \int_t^\infty \lambda_1 e^{-\lambda_1 x} [-e^{-\lambda_2 y}]_t^x dx \\
&= \int_t^\infty \lambda_1 e^{-\lambda_1 x} (e^{-\lambda_2 t} - e^{-\lambda_2 x}) dx \\
&= e^{-\lambda_2 t} \int_t^\infty \lambda_1 e^{-\lambda_1 x} dx - \int_t^\infty \lambda_1 e^{-(\lambda_1 + \lambda_2)x} dx \\
&= e^{-\lambda_2 t} [-e^{-\lambda_1 x}]_t^\infty - \left[-\frac{\lambda_1}{\lambda_1 + \lambda_2} e^{-(\lambda_1 + \lambda_2)x} \right]_t^\infty \\
&= e^{-\lambda_2 t} e^{-\lambda_1 t} - \frac{\lambda_1}{\lambda_1 + \lambda_2} e^{-(\lambda_1 + \lambda_2)t} \\
&= e^{-(\lambda_1 + \lambda_2)t} \frac{\lambda_2}{\lambda_1 + \lambda_2}.
\end{aligned}$$

hence we have that,

$$\begin{aligned}
P(N = 2, U > t) &= \frac{\lambda_2}{\lambda_1 + \lambda_2} e^{-(\lambda_1 + \lambda_2)t} \\
&= P(N = 2)P(U > t).
\end{aligned}$$

Thus we have that

$$P(N = 1, U > t) = P(N = 1)P(U > t)$$

and

$$P(N = 2, U > t) = P(N = 2)P(U > t)$$

Since these cover all the possible values of the random variable N then N and U are independent random variables.

4. We want to show that $P(W > t \mid N = 1) = e^{\lambda_2 t}$ and $P(W > t \mid N = 2) = e^{\lambda_1 t}$ for $t \geq 0$

Notice that for $X_1 - X_2 < 0$, then $W = X_2 - X_1$. Thus given that $N = 1$ we have that $W = X_2 - X_1$. In particular, we have that

$$\begin{aligned}
P(W > t \mid N = 1) &= P(X_2 - X_1 > t \mid X_1 < X_2) \\
&= \frac{P(X_2 - X_1 > t, X_1 < X_2)}{P(X_1 < X_2)} \\
&= \frac{P(X_2 > X_1 + t, X_1 < X_2)}{P(X_1 < X_2)}
\end{aligned}$$

Thus we get the following:

$$\begin{aligned}
P(X_2 - X_1 > t, X_1 < X_2) &= \int_0^\infty \int_{x+t}^\infty \lambda_1 e^{-\lambda_1 x} \lambda_2 e^{-\lambda_2 y} dy dx \\
&= \int_0^\infty \lambda_1 e^{-\lambda_1 x} [-e^{-\lambda_2 y}]_{x+t}^\infty dx \\
&= \int_0^\infty \lambda_1 e^{-\lambda_1 x} (0 + e^{-\lambda_2(x+t)}) dx \\
&= \int_0^\infty \lambda_1 e^{-\lambda_1 x} e^{-\lambda_2(x+t)} dx \\
&= e^{-\lambda_2 t} \int_0^\infty \lambda_1 e^{-(\lambda_1 + \lambda_2)x} dx \\
&= e^{-\lambda_2 t} \int_0^\infty \lambda_1 e^{-(\lambda_1 + \lambda_2)x} dx \\
&= e^{-\lambda_2 t} \int_0^\infty \lambda_1 e^{-(\lambda_1 + \lambda_2)x} dx \\
&= e^{-\lambda_2 t} \left[\frac{-\lambda_1}{\lambda_1 + \lambda_2} e^{-(\lambda_1 + \lambda_2)x} \right]_0^\infty \\
&= e^{-\lambda_2 t} \left[\frac{\lambda_1}{\lambda_1 + \lambda_2} \right] \\
&= e^{-\lambda_2 t} \frac{\lambda_1}{\lambda_1 + \lambda_2}.
\end{aligned}$$

We also know that

$$P(N = 1) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

Hence we have that our initial conditional probability becomes

$$P(W > t \mid N = 1) = e^{-\lambda_2 t}.$$

Consider now the case where $N = 2$, thus we know that in that event we have that $X_1 \geq X_2$, then $W = X_1 - X_2$. Hence,

$$P(W > t \mid N = 2) = P(X_1 - X_2 > t \mid X_1 \geq X_2) = \frac{P(X_1 > X_2 + t, X_1 \geq X_2)}{P(N = 2)}.$$

We then have the following:

$$\begin{aligned}
P(X_1 > X_2 + t, X_1 \geq X_2) &= \int_0^\infty \int_{y+t}^\infty \lambda_2 e^{-\lambda_2 y} \lambda_1 e^{-\lambda_1 x} dx dy \\
&= \int_0^\infty \lambda_2 e^{-\lambda_2 y} e^{-\lambda_1(y+t)} dy \\
&= e^{-\lambda_1 t} \frac{\lambda_2}{\lambda_1 + \lambda_2}.
\end{aligned}$$

We already know that

$$P(N = 2) = \frac{\lambda_2}{\lambda_2 + \lambda_1}$$

thus

$$P(W > t \mid N = 2) = e^{-\lambda_1 t}.$$

Exercise 2

I

Consider a bivariate random variable $X = (X_1, X_2)$. We say the random vector X follows the bivariate normal distribution $\mathcal{N}(\mu, \Sigma)$, with $\mu = [\mu_1, \mu_2]'$ and $\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{bmatrix}$, a symmetric positive definite matrix denoted by

$$\begin{bmatrix} X_1 & X_2 \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{bmatrix} \right),$$

if and only if its probability density function is expressed as:

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi\sqrt{\det(\Sigma)}} \exp \left(-\frac{1}{2}(x - \mu)' \Sigma^{-1} (x - \mu) \right)$$

1. To show that the pdf of the random vector can be expressed in the given form we proceed by calculating each of the individual components.

We have that

$$\det \Sigma = (\sigma_1 \sigma_2)^2 - \sigma_{12} \sigma_{21} = (\sigma_1 \sigma_2)^2 - \sigma_{12}^2 = (\sigma_1^2 \sigma_2^2) \left(1 - \frac{\sigma_{12}^2}{(\sigma_1 \sigma_2)^2} \right) = (\sigma_1^2 \sigma_2^2) (1 - \rho^2)$$

since Σ is a symmetric matrix, with $\rho = \sqrt{\frac{\sigma_{12}^2}{(\sigma_1 \sigma_2)^2}}$

Next we have that

$$\Sigma^{-1} = \frac{1}{\sigma_1^2 \sigma_2^2 (1 - \rho^2)} \begin{bmatrix} \sigma_2^2 & -\sigma_{12} \\ -\sigma_{12} & \sigma_1^2 \end{bmatrix}$$

we also have that $x - \mu = \begin{bmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \end{bmatrix}$ finally we have that

$$\begin{aligned} [x - \mu]' \Sigma^{-1} [x - \mu] &= \frac{1}{\sigma_1^2 \sigma_2^2 (1 - \rho^2)} \left(((x_1 - \mu_1) \sigma_2^2 - (x_2 - \mu_2) \sigma_{12}) (x_1 - \mu_1) \right. \\ &\quad \left. + \frac{1}{\sigma_1^2 \sigma_2^2 (1 - \rho^2)} \left(-(x_1 - \mu_1) \sigma_{12} + (x_2 - \mu_2) \sigma_1^2 \right) (x_2 - \mu_2) \right) \end{aligned}$$

Now plugging each of these into our original f_{X_1, X_2} we get the following:

$$f_{X_1, X_2}(x_1, x_2) = f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi \sigma_1 \sigma_2 \sqrt{1 - \rho^2}}.$$

$$e^{\left(-\frac{1}{2(1-\rho^2)} \left[\frac{(x_1 - \mu_1)^2}{\sigma_1^2} - 2\rho \frac{(x_1 - \mu_1)}{\sigma_1} \frac{(x_2 - \mu_2)}{\sigma_2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2} \right] \right)}$$

As required.

2. The marginal pdf of X_1 is given as

$$\begin{aligned}
f_{X_1}(x_1) &= \int_{-\infty}^{\infty} f_{X_1, X_2}(x_1, x_2) dx_2 \\
&= \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x_1-\mu_1)}{\sigma_1\sigma_2}(x_2-\mu_2) + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]\right) dx_2 \\
&= \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left(-\frac{(x_1-\mu_1)^2}{2\sigma_1^2}\right) \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2(1-\rho^2)} \left[-2\rho\frac{(x_1-\mu_1)}{\sigma_1\sigma_2}(x_2-\mu_2) + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]\right) dx_2 \\
&= \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left(-\frac{(x_1-\mu_1)^2}{2\sigma_1^2}\right) \sqrt{2\pi\sigma_2^2(1-\rho^2)} \\
&= \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left(-\frac{(x_1-\mu_1)^2}{2\sigma_1^2}\right)
\end{aligned}$$

Similarly we find that the marginal pdf of X_2 is given as

$$\begin{aligned}
f_{X_2}(x_2) &= \int_{-\infty}^{\infty} f_{X_1, X_2}(x_1, x_2) dx_1 \\
&= \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x_1-\mu_1)}{\sigma_1\sigma_2}(x_2-\mu_2) + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]\right) dx_1 \\
&= \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left(-\frac{(x_2-\mu_2)^2}{2\sigma_2^2}\right) \int_{-\infty}^{\infty} \frac{1}{\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x_1-\mu_1)}{\sigma_1\sigma_2}(x_2-\mu_2) + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]\right) dx_1 \\
&= \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left(-\frac{(x_2-\mu_2)^2}{2\sigma_2^2}\right) \sqrt{2\pi\sigma_1^2(1-\rho^2)} \\
&= \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left(-\frac{(x_2-\mu_2)^2}{2\sigma_2^2}\right)
\end{aligned}$$

3. We now want to find the conditional pdf of X_1 given that $X_2 = x_2$ and the conditional pdf of X_2 given $X_1 = x_1$

We find

$$f_{X_1|X_2=x_2}(x_1 | x_2) = \frac{f_{X_1, X_2}(x_1, x_2)}{f_{X_2}(x_2)}$$

.

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)} \left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x_1-\mu_1)}{\sigma_1\sigma_2}(x_2-\mu_2) + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]}$$

Thus we have that $f_{X_1|X_2=x_2}(x_1 | x_2)$ can be given as

$$\frac{\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)} \left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x_1-\mu_1)}{\sigma_1\sigma_2}(x_2-\mu_2) + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]}}{\frac{1}{\sqrt{2\pi\sigma_2^2}} e^{-\frac{(x_2-\mu_2)^2}{2\sigma_2^2}}}$$

Similarly we compute the conditional probability of X_2 , given that $X_1 = x_1$ as

$$\frac{f_{X_1, X_2}(x_1, x_2)}{f_{X_1}(x_1)} = \frac{1}{\sqrt{2\pi\sigma_2^2(1-\rho^2)}} e^{-\frac{1}{2\sigma_2^2(1-\rho^2)}(x_2 - (\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x_1 - \mu_1)))^2}$$

4. Notice, from the definition of the joint pdf, that when $\rho = 0$ the term in the exponential of the pdf becomes

$$e^{-\frac{1}{2}\left[\left(\frac{x_1 - \mu_1}{\sigma_1}\right)^2 + \left(\frac{x_2 - \mu_2}{\sigma_2}\right)^2\right]} = e^{-\frac{1}{2}\left(\frac{x_1 - \mu_1}{\sigma_1}\right)^2} e^{-\frac{1}{2}\left(\frac{x_2 - \mu_2}{\sigma_2}\right)^2}$$

Thus our joint pdf can be written as

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{\sigma_1\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_1 - \mu_1}{\sigma_1}\right)^2} \frac{1}{\sigma_2\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_2 - \mu_2}{\sigma_2}\right)^2} = f_{X_2}(x_2)f_{X_1}(x_1)$$

Thus when $\rho = 0$ the random variables are independent.

II

Let (X_1, X_2) be a bivariate random variable with joint pdf given as

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi} e^{-(x_1^2 + x_2^2)/2} (1 + x_1 x_2 e^{-(x_1^2 + x_2^2 - 2)/2}), \text{ for } x_1 \in \mathbb{R} \text{ and } x_2 \in \mathbb{R}$$

1. We want to show that the pdf is a probability distribution. Notice that since the exponential function is a positive function then the terms in the exponent will always be positive regardless of the value of x_1 and x_2 . Now notice that we have a term that contains $x_1 x_2$ and depending on the values of x_1 and x_2 that product could be either positive or negative. If the product is positive then we are done, since all the other terms in the pdf are positive too. However in the case that they are negative then exponential factor of the product $x_1 x_2$ gets very close to 0 such that the entire term gets a very small negative value between 0 and 1. (This is because the exponential term decays/grows at a faster rate than the linear functions x_1 and x_2) And so our joint pdf is always positive, regardless of the values of x_1 and x_2 .

Now We need to show that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 = 1$$

Given:

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi} e^{-(x_1^2 + x_2^2)/2} (1 + x_1 x_2 e^{-(x_1^2 + x_2^2 - 2)/2})$$

We first split the integral:

$$I = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-(x_1^2 + x_2^2)/2} dx_1 dx_2 + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{2\pi} x_1 x_2 e^{-(x_1^2 + x_2^2 - 2)/2} dx_1 dx_2$$

Notice that the first term in the sum I is the bivariate standard normal distribution: Thus we have that

$$I_1 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-(x_1^2+x_2^2)/2} dx_1 dx_2 = 1$$

The second term evaluates to zero due to symmetry:

$$I_2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{2\pi} x_1 x_2 e^{-(x_1^2+x_2^2-2)/2} dx_1 dx_2 = 0$$

Hence

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 = 1$$

Thus, $f_{X_1, X_2}(x_1, x_2)$ is a valid probability distribution.

2. We find and identify the marginal pdfs of X_1 as follows:

$$f_{X_1}(x_1) = \frac{1}{2\pi} e^{-\frac{x_1^2}{2}} \left(\int_{-\infty}^{\infty} e^{-\frac{x_2^2}{2}} dx_2 + x_1 \int_{-\infty}^{\infty} x_2 e^{-\frac{x_1^2+x_2^2}{2}} e^{-\frac{x_2^2-2}{2}} dx_2 \right)$$

Notice that the first integral in the expression above is the Gaussian integral which is $\sqrt{2\pi}$, and the second integral evaluates to zero due to symmetry. Hence, the marginal p.d.f. of X_1 is:

$$f_{X_1}(x_1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x_1^2}{2}}$$

Similarly the marginal pdf of X_2 is as follows:

$$f_{X_2}(x_2) = \frac{1}{2\pi} e^{-\frac{x_2^2}{2}} \left(\int_{-\infty}^{\infty} e^{-\frac{x_1^2}{2}} dx_1 + x_2 \int_{-\infty}^{\infty} x_1 e^{-\frac{x_1^2+x_2^2}{2}} e^{-\frac{x_1^2-2}{2}} dx_1 \right)$$

And with the same arguments as before we have that

$$f_{X_2}(x_2) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x_2^2}{2}}$$

Both of these marginal pdfs are the standard normal distribution in one variable.

3. No f_{X_1, X_2} is not a bivariate normal distribution since if you compare the two pdfs we see the presence of a -2 which is not present in pdf of the bivariate normal. The pdf given cannot also be manipulated to look like the pdf of the bivariate normal distribution hence it is not a bivariate normal distribution.

Exercise 3

Let $(X_1, X_2, X_3, \dots, X_n)$ be n random variables that that are identical and independently distributed each with probability density function:

$$f(x | \theta) = \theta(\theta + 1)x^{\theta-1}(1 - x) \quad 0 < x < 1, \theta > 0 \quad (5)$$

I. Estimation

1. We want to show that $f(x | \theta)$ is a probability density function. Notice that since $\theta > 0$ and $0 < x < 1$ then $f(x | \theta)$ is always greater than 0. Now we want to show that

$$\begin{aligned}\int_0^1 \theta(\theta + 1)x^{\theta-1}(1 - x)dx &= \theta(\theta + 1) \left(\int_0^1 x^{\theta-1}dx - \int_0^1 x^{\theta}dx \right) \\ &= \theta(\theta + 1) \left(\frac{x^{\theta}}{\theta} \Big|_0^1 - \frac{x^{\theta+1}}{\theta + 1} \Big|_0^1 \right) \\ &= \theta(\theta + 1) \left(\frac{1}{\theta} - \frac{1}{\theta + 1} \right) \\ &= \theta(\theta + 1) \left(\frac{\theta + 1 - \theta}{\theta(\theta + 1)} \right) \\ &= 1\end{aligned}$$

Thus f is a probability distribution.

2. We want to use the method of moments to find an estimator of θ

We know that the first empirical moment is given as

$$m_1 = \frac{1}{n} \sum_{i=1}^n X_i$$

Now using the method of moment estimators, we know that the estimator of θ is the solution to the equation

$$m_1 = E(X) \tag{6}$$

But we already know that

$$\begin{aligned} E(X) &= \int_0^1 x\theta(\theta+1)x^{\theta-1}(1-x)dx \\ &= \theta(\theta+1) \int_0^1 x^{\theta}(1-x)dx \\ &= \theta(\theta+1) \left(\frac{x^{\theta+1}}{\theta+1} \Big|_0^1 - \frac{x^{\theta+2}}{\theta+2} \Big|_0^1 \right) \\ &= \theta(\theta+1) \left(\frac{1}{\theta+1} - \frac{1}{\theta+2} \right) \\ &= \theta(\theta+1) \left(\frac{\theta+2-\theta-1}{(\theta+1)(\theta+2)} \right) \\ &= \frac{\theta}{\theta+2} \end{aligned}$$

But we also know that

$$m_1 = \overline{X}_n$$

thus

$$\overline{X}_n = \frac{\theta}{\theta+2}$$

$$\theta = \overline{X}_n\theta + 2\overline{X}_n$$

$$\theta = \frac{2\overline{X}_n}{1 - \overline{X}_n}$$

And so an estimator of θ by the method of moments is given as

$$\hat{\theta} = \frac{2\overline{X}_n}{1 - \overline{X}_n}$$

3. To find the maximum likelihood estimator we proceed as follows:

We know that the likelihood function is given as

$$\begin{aligned}
L(\theta) &= f(x, y, \dots, x_n) \\
&= \prod_{i=1}^n f(x_i) \quad \text{Since the } X_i\text{s are independent} \\
&= \prod_{i=1}^n [\theta(\theta+1)x_i^{\theta-1}(1-x_i)] \\
&= \theta^n(\theta+1)^n \prod_{i=1}^n x_i^{\theta-1}(1-x_i)
\end{aligned}$$

we now take the log-likelihood of $L(\theta)$ to get:

$$\begin{aligned}
\log(L(\theta)) &= \log(\theta^n(\theta+1)^n \prod_{i=1}^n x_i^{\theta-1}(1-x_i)) \\
&= n \log(\theta) + n \log(\theta+1) + (\theta-1) \sum_{i=1}^n \log x_i + \sum_{i=1}^n \log(1-x_i)
\end{aligned}$$

We now want to find the maximum value of $\log(L(\theta))$ so we take the derivative and set it to 0 as follows:

$$\begin{aligned}
[\log(L(\theta))]' &= \frac{n}{\theta} + \frac{n}{\theta+1} + \sum_{i=1}^n \log x_i \\
&= \frac{2n\theta+n}{\theta(\theta+1)} + \sum_{i=1}^n \log x_i
\end{aligned}$$

Setting it to 0 we get:

$$\begin{aligned}
0 &= \frac{2n\theta+n}{\theta(\theta+1)} + \sum_{i=1}^n \log x_i \\
-\frac{1}{n} \sum_{i=1}^n \log x_i &= \frac{2\theta+1}{\theta(\theta+1)} \\
\bar{L}_n &= \frac{2\theta+1}{\theta(\theta+1)} \quad \text{where } \bar{L}_n = -\frac{1}{n} \sum_{i=1}^n \log x_i < 0 \\
0 &= \theta^2 \bar{L}_n + \theta(\bar{L}_n + 2) + 1
\end{aligned}$$

Thus we get that an estimator $\hat{\theta}$, of θ is

$$\hat{\theta} = \frac{-(\bar{L}_n - 2) \pm \sqrt{(\bar{L}_n - 2)^2 + 4\bar{L}_n}}{2\bar{L}_n}$$

but since we are taking the maximum likelihood estimator, we need $\hat{\theta}$ to be maximum thus

$$\hat{\theta} = \frac{-(\bar{L}_n - 2) + \sqrt{(\bar{L}_n - 2)^2 + 4\bar{L}_n}}{2\bar{L}_n}$$

II. Simulation with Python

Notice that we can rewrite equation 5 as

$$f(x | \theta) = \left(\frac{1}{\frac{1}{\theta(\theta+1)}} \right) x^{\theta-1} (1-x)^{2-1}$$

comparing this with the definition of the p.d.f of a random variable following a Beta distribution we see that each X_i follows a Beta distribution with parameters $\alpha = \theta$, $\theta > 0$ and $\beta = 2$.

Notice that

$$\begin{aligned} B(\theta, 2) &= \frac{\Gamma(\theta)\Gamma(2)}{\Gamma(\theta+2)} \\ &= \frac{\Gamma(\theta)\Gamma(2)}{\Gamma((\theta+1)+1)} \\ &= \frac{\Gamma(\theta)\Gamma(2)}{(\theta+1)\Gamma(\theta+1)} \\ &= \frac{\Gamma(\theta)\Gamma(2)}{\theta(\theta+1)\Gamma(\theta)} \\ &= \frac{\Gamma(2)}{\theta(\theta+1)} \end{aligned}$$

Now from the definition of the Γ function, we know that $\Gamma(2) = \int_0^\infty u e^{-u} du = 1$ hence we have that $B(\theta, 2) = \frac{\Gamma(2)}{\theta(\theta+1)} = \frac{1}{\theta(\theta+1)}$